Incomplete reversibility of fixation-induced changes in postmortem human brain WM MRI following rehydration

Lennart Bedarf[1,2], Dominique Neuhaus[1,2], Eva Scheurer[1,2], Claudia Lenz[1,2]

[1]Institute of Forensic Medicine, Department of Biomedical Engineering, University of Basel, Switzerland
[2]Institute of Forensic Medicine, Health Department Basel-Stadt, Basel, Switzerland

University of Basel

Swiss National Science Foundation

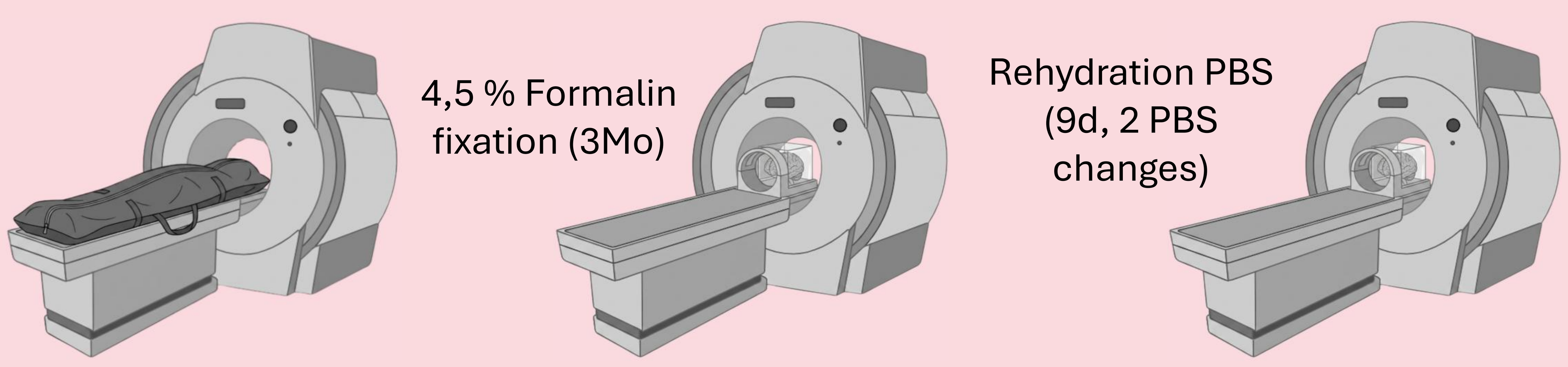
Background and Objective

- Validation of MRI biomarkers: based on formalin-fixed ex situ imaging and histology
- Fixation: alters relaxation properties and tissue anisotropy[1-4]
- Rehydration: suspected to partially reverse fixation effects → irreversible protein cross-linking suggests incomplete reversibility[4-5]

Objective: Assess effects of formalin fixation and rehydration on MRI derived measures in human brain WM

Methods

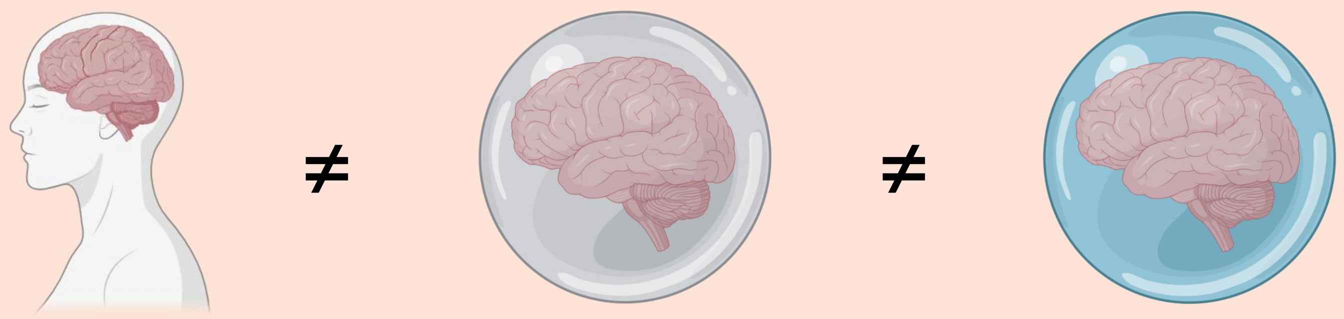
- **Subjects & MRI:**
  - Two human brains scanned on Siemens Prisma 3T



- **MRI Protocol:**
  - MP2RAGE: structural imaging and tissue segmentation
  - Multi-echo GRE (12 echoes): R_2^* mapping
  - Multi-shell DTI (b = 0, 1000, 2000, 3000 s/mm^2)
- **Image processing:**
  - HD-BET[6] - brain extraction, FSL FAST[7-10] - segmentation, FSL eddy_correct[7-9], FSL DTIFIT[7-9] - mean diffusivity (MD) , fractional anisotropy (FA), axial diffusivity (AD), radial diffusivity (RD)
- **Analysis:**
  - Fiber angles derived from DTI V1, binned 18 x 5° (0° - 90°)
  - R_2^* calculated voxel-wise, data normalised and registered
  - R_2^* anisotropy index AI = $(R_2^{*\,max} - R_2^{*\,min})/(R_2^{*\,max} + R_2^{*\,min})$

Conclusion

- **Fixation and rehydration:**
  - substantial alterations in MRI-derived measures
  - heterogeneous & partial reversibility



Comparison of in vivo MRI to ex situ measurements for biomarker validation is limited

Results and Discussion

- Formalin fixation alters WM microstructure
- Fixation: FA ↓, MD ↓, AD ↓, RD ↓, AI ↓, R_2^* ↑
- Rehydration: FA ↓ ,MD ↑, AD ↑, RD ↑, AI ↑, R_2^* ↓
- Angular dependency: in situ > rehydrated > fixed

Table 1: Diffusion parameters FA, MD, AD and RD, as well as R_2^* and anisotropy index across the three tissue conditions.

	In situ	PM ex situ formalin-fixed	PM ex situ rehydrated
FA	0.5 ± 0.0	0.4 ± 0.1	0.3 ± 0.0
MD [10^{-5} mm^2/s]	16.2 ± 6. 5	13.0 ± 1.3	15.3 ± 2.8
AD [10^{-5} mm^2/s]	24.8 ± 9.8	17.5 ± 2.2	19.9 ± 4.0
RD [10^{-5} mm^2/s]	12.1 ± 4.7	11.0 ± 1.1	13.0 ± 2.3
R_2^* [s^{-1}]	27.8 ± 4.5	32.4 ± 0.4	28.4 ± 0.4
AI [%]	9.3 ± 3.3	1.5 ± 0.0	2.8 ± 0.4

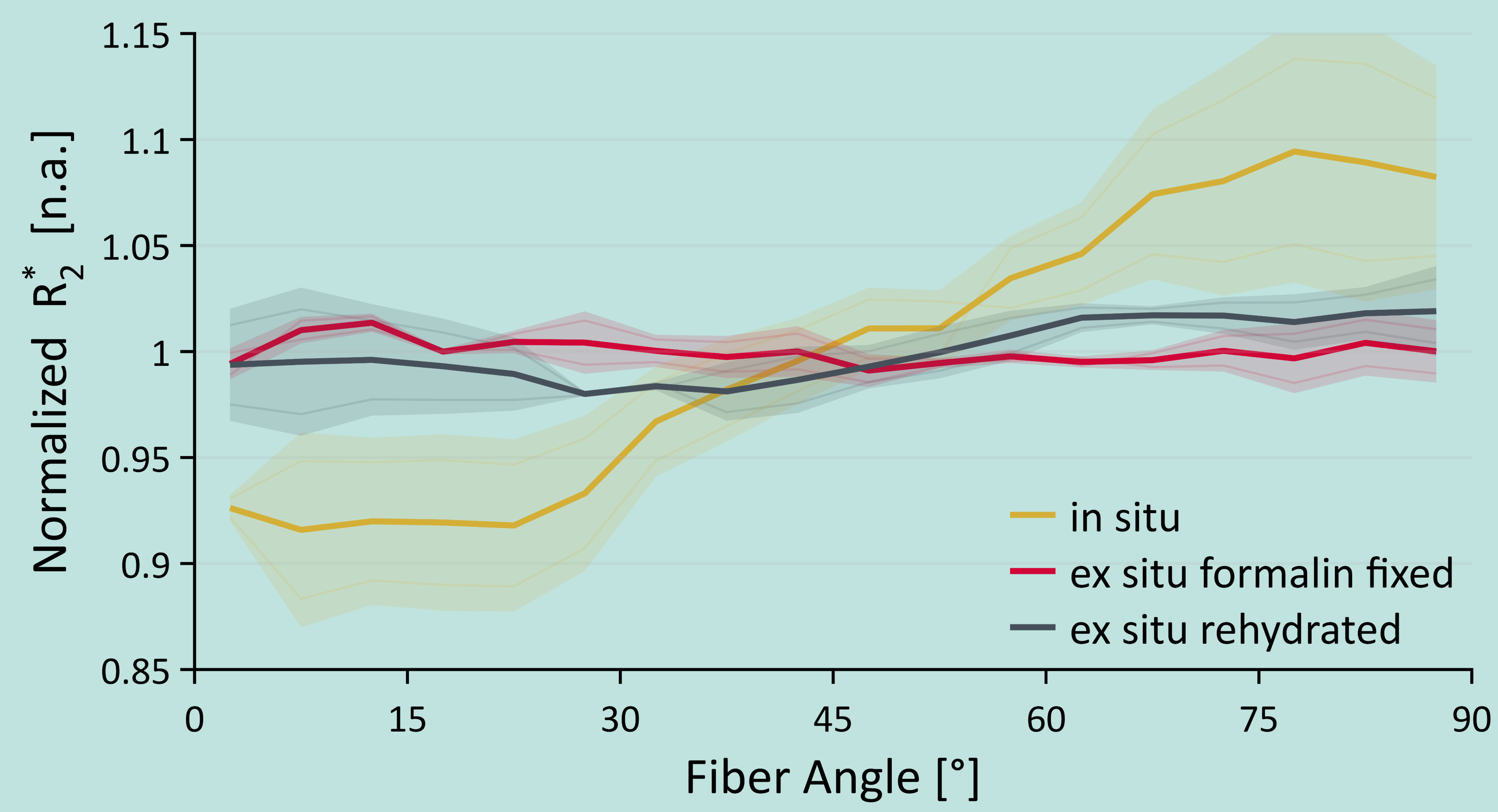


Figure 1: Normalized R_2^* values across the three tissue conditions over the calculated tissue fiber angle.

- **Formalin fixation:**
  - Diffusivity, FA and R_2^* orientation dependency: ↓
- **Rehydration:**
  - partial recovery of MD, AD, AI and R_2^*
  - FA remained reduced, RD exceeded in situ
  - → Tissue properties not fully restored

Key Message: Fixation effects are partly reversible, but not fully recoverable through rehydration

- Osmotic shifts & water distribution may cause apparent («pseudo-») recovery
- Fixation-induced changes should be considered when validating in vivo biomarkers using ex situ MRI and histology

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